

JOSHUA TOMLIN

Mathematician & Software Developer

+61 415 626 592 | joshuatomlin@gmail.com | Adelaide, Australia | joshtomlin.github.io

SUMMARY

Pure mathematician with a PhD in Mathematics specialising in 4-dimensional topology and gauge theory. Experienced in conducting independent and collaborative research, formulating proofs and communicating complex mathematics. Also an experienced programmer, having built software in C++, Python and JavaScript. Currently working as a Data Scientist, writing code in SQL, R and Python.

EDUCATION

Doctor of Philosophy - Mathematics. *University of Adelaide.* Jan 2022 - Oct 2025

- Researched Seiberg–Witten theory of 4-manifold families, earning the Dean's Commendation for thesis excellence and the George Fraser Supplementary Scholarship.
- Derived an original connected sum formula for 4-manifold families that applies in the most general setting. Produced two independently authored papers.

Bachelor of Philosophy (Honours) - Mathematics. *Australian National University.* Jan 2016 - Dec 2019

- Achieved a perfect 7.0 GPA, First Class Honours and an overall mark of 93%.
- Researched tight Dehn surgery on contact 3-manifolds with Prof Joan Licata.
- Took computer science electives in programming and SML, with experience in Python, Java and C++.

Remote Coursework. *Northwestern University, Evanston, IL.* Sep 2020 - Sep 2021

- Accepted into the PhD program and passed preliminary examinations. Completed courses in topology, analysis, probability and stochastic differential equations. Withdrew after one year to pursue a PhD in Australia.

EXPERIENCE

Data Scientist. *Amplar Home Health Hospital.* Jan 2026 - Present

- Delivered recurring KPI reporting within strict deadlines through automation using SQL, R and Excel.
- Tested data pipelines, identifying quality issues and validating results before release to stakeholders.
- Documented code and reporting methodologies to ensure processes were transparent and maintainable.
- Developed reports and dashboards in Power BI, Amazon QuickSight and SSRS.

Associate Lecturer. *University of Adelaide.* Jul 2025 - Dec 2025

- Conducted research on an original connected sum formula for 4-manifold families. Produced two significant new results and two independently authored papers.
- Collaborated with a colleague to investigate further applications of my research relating to exotic diffeomorphisms and mapping class groups, culminating in a third paper.
- Coordinated a third-year complex analysis course and managed curriculum delivery and improvement.
- Led the development of a new algebra course. Designed learning outcomes, curated topics and wrote assessments.

Workshop Demonstrator. *University of Adelaide, Australian National University.* Mar 2018 - Jul 2025

- Delivered tutorials in Linear Algebra, Calculus and Discrete Mathematics, guiding students through mathematical problem solving and explaining complex concepts clearly.
- Refined teaching practices over six years in response to student feedback, resulting in a consistent 95% satisfaction rating from student surveys.

PUBLICATIONS

D. Baraglia, J. Tomlin, *Exotic diffeomorphisms of 4-manifolds with $b_+ = 2$* , 2025. arXiv: 2511.13976.

J. Tomlin, *A general connected sum formula for the families Seiberg–Witten invariant*, 2025. arXiv: 2510.17141.

J. Tomlin, *A general connected sum formula for the families Bauer–Furuta invariant*, 2025. arXiv: 2510.14201.

PROGRAMMING PROJECTS

Bridge Lab. *C++* | *WebAssembly* | *JavaScript*.

2026 - Present

- Built a solver for analysing declarer play in contract bridge when the defenders' cards are unknown.
- Designed the solver to sample possible defender layouts and use alpha-mu search to find one strategy that works across as many of them as possible.
- Implemented the bridge rules, deal sampling and search engine in C++. The engine compiles to WebAssembly and uses a JavaScript interface for deal entry, interactive play and analysis in the browser.
- Researched and implemented various optimisations to reduce the search space and improve performance.

JBLD. *React* | *JavaScript* | *PWA*.

2026 - Present

- Built a mobile training and analysis app for blindfolded Rubik's Cube solving that records timestamped moves from Bluetooth smart cubes.
- The app reconstructs each recorded solve into the individual algorithms used. It identifies pauses, timing losses and execution errors and generates a recommended solution for comparison.
- Built the app as a local-first React PWA. Bluetooth capture, cube replay and analysis run in the browser. It works offline and stores solve history and personal algorithms locally.

Love Letter. *TypeScript* | *React*.

2026 - Present

- Built a playable two-player browser version of Love Letter with an explainable bot that reasons about an opponent's hidden hand.
- The bot updates a probability distribution from public actions. It scores each legal play and tests actions through sampled games. An in-game panel shows its assumptions, action values and final choice.
- Built the game rules and bot in TypeScript and the browser interface with React.

ADDITIONAL EXPERIENCE

Faculty Seminars. *University of Adelaide*.

Jan 2022 - Oct 2025

Prepared and delivered talks on 4-manifold topology, Seiberg–Witten theory, contact geometry and other advanced mathematical topics.

Bulletin Editor. *International Bridge Press Association*.

Oct 2023 - Dec 2025

Wrote and edited articles about contract bridge for an international audience, making technical material accessible to players of different skill levels.

Youth Coordinator. *South Australian Bridge Federation*.

Jan 2024 - Present

Led initiatives to grow youth bridge participation in South Australia, designing outreach activities and coordinating training sessions that doubled engagement across the state.